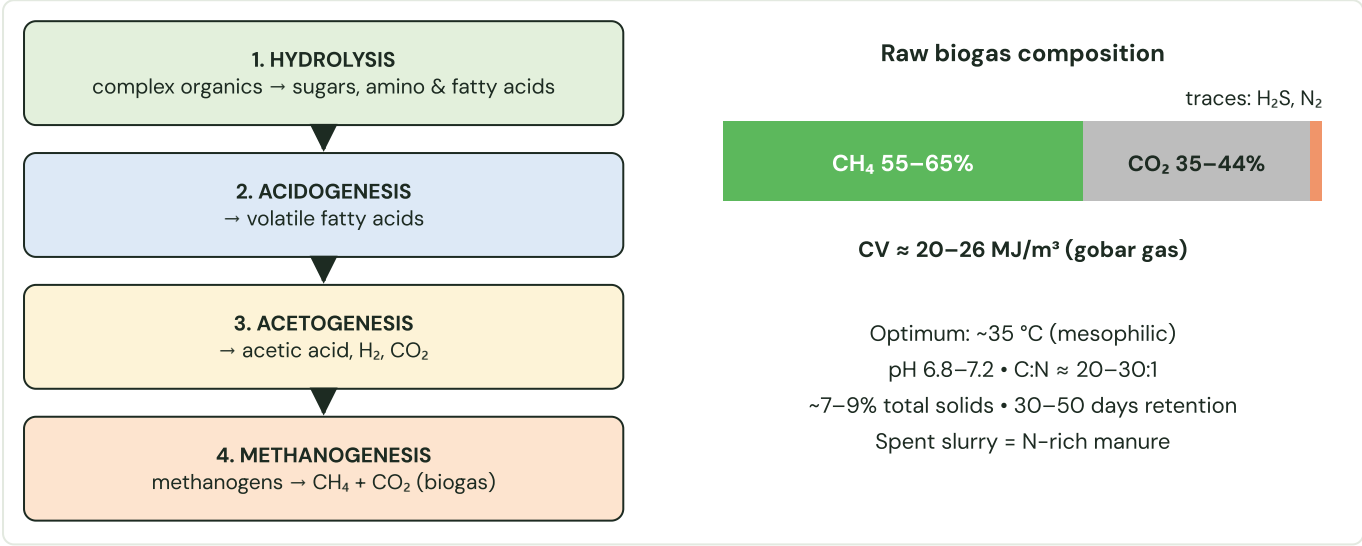
UNIT 7 · TOPIC 9

Biogas — Digestion, Plant Types, Properties, Uses and Distribution

Syllabus topic: Types of biogas plants; biogas properties, uses and distribution

Anaerobic digestion → biogas



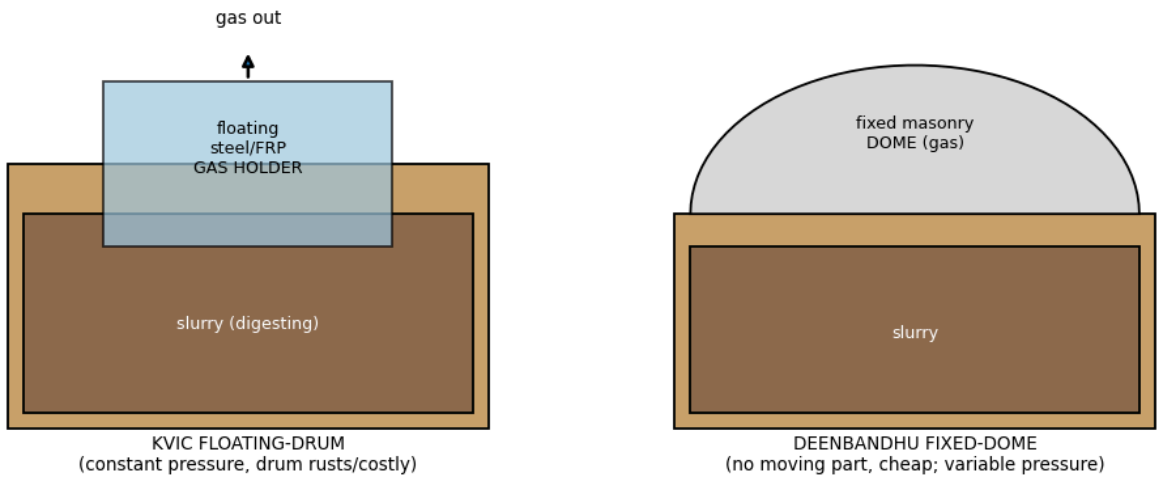
Anaerobic digestion stages and raw biogas composition (redrawn — the source figure's arrows ran from methanogenesis back to hydrolysis)

1. **HYDROLYSIS** — complex molecules → sugars, amino and fatty acids.
2. **ACIDOGENESIS** — → volatile fatty acids.
3. **ACETOGENESIS** — → acetic acid, H_2, CO_2.
4. **METHANOGENESIS** — methanogens → CH_4 + CO_2.

Condition	Optimum
Temperature	~35 °C (mesophilic)
pH	6.8–7.2 (near neutral)
C:N ratio	~20–30:1
Total solids	~7–9%
Retention time	30–50 days

- **BIOGAS (gobar gas) ≈ 55–65% METHANE + 35–44% CO_2**, traces of H_2S and N_2; CV ~**20–26 MJ/m³**.
- The spent slurry is a good N-rich MANURE.

Types of biogas plant



KVIC floating-drum vs Deenbandhu fixed-dome biogas plants

Plant	Construction	Advantages	Drawbacks
FLOATING-DRUM (KVIC)	Inverted steel/FRP gas holder floats on the slurry in a masonry digester	Gas at CONSTANT pressure; shows the stored volume	Steel drum corrodes; costlier
FIXED-DOME (Janata / DEENBANDHU)	Fixed masonry dome stores gas above the slurry	NO moving part; cheaper and durable	Pressure VARIES; needs skilled, gas-tight masonry
Bag/balloon; community/institutional plants	Portable or large shared digesters	—	—

Feedstock: cattle dung, night soil, poultry litter, crop residue, kitchen/agro-waste.

Properties, uses and distribution

- Clean, smokeless flame.
- **Uses:** COOKING (biogas stove), LIGHTING (mantle lamp), DUAL-FUEL IC engines/gensets and pumps; when purified and compressed, CBG vehicle fuel.
- **Distribution:** low-pressure gas by pipeline from the dome/holder to the point of use (short distances on-farm); for wider use it is upgraded and bottled as CBG.
- The slurry returns to the field as manure — a twin energy-plus-fertiliser benefit.

Updated information — biogas/CBG in India (as given in the source notes, July 2026)

Item	Status
CBG under SATAT	132 Compressed Bio-Gas plants commissioned, ~920 tonnes/day (2026); >1,200 letters of intent but <400 plants operational (execution gap)
SATAT target	5,000 CBG plants (~15 MMT/yr)
Support	MNRE central financial assistance up to ₹10 crore/plant; 15-year OMC offtake; priority-sector lending
CBG quality (IS 16087:2016)	Minimum **90% methane**, compressed to ~**250 bar**; raw biogas (55–65% CH_4) is upgraded to ~90–98% CH_4
Cooperatives	India–Japan initiative to build 1,000 cooperative biogas plants by 2030

Practice MCQs

Q106. Biogas is produced by:

(a) Aerobic composting
(b) Anaerobic digestion of organic matter
(c) Combustion
(d) Gasification

Q107. The four stages of anaerobic digestion end with:

(a) Hydrolysis
(b) Acidogenesis
(c) Acetogenesis
(d) Methanogenesis

Q108. Methane content of raw biogas is about:

(a) 10–20%
(b) 30–40%
(c) 55–65%
(d) 90–95%

Q109. The major non-combustible component of raw biogas is:

(a) CO_2 (35–44%)
(b) N_2
(c) O_2
(d) H_2

Q110. Optimum temperature for mesophilic digestion is about:

(a) 10 °C
(b) 35 °C
(c) 60 °C
(d) 90 °C

Q111. The ideal C:N ratio for biogas feedstock is about:

(a) 5:1
(b) 20–30:1
(c) 60:1
(d) 100:1

Q112. The KVIC biogas plant uses a:

(a) Fixed masonry dome
(b) Floating steel/FRP gas holder
(c) Balloon
(d) Rock bed

Q113. An advantage of the floating-drum plant is:

(a) No moving parts
(b) Constant gas pressure
(c) Lowest cost
(d) No corrosion

Q114. The Deenbandhu plant is a:

(a) Floating drum
(b) Fixed-dome plant (no moving part, cheaper)
(c) Portable bag
(d) Community digester only

Q115. A drawback of the fixed-dome plant is:

(a) Corrosion of drum
(b) Variable gas pressure
(c) Moving parts
(d) High steel cost

Q116. The spent slurry from a biogas plant is valuable as:

(a) Fuel
(b) Nitrogen-rich manure
(c) Pesticide
(d) Cement

Q123. Compressed Bio-Gas (CBG) per IS 16087:2016 must contain minimum methane of:

(a) 50%
(b) 75%
(c) 90%
(d) 99%

Q124. CBG is compressed to a pressure of about:

(a) 5 bar
(b) 50 bar
(c) 250 bar
(d) 1000 bar

Q125. The Indian scheme promoting CBG plants is:

(a) SATAT
(b) FAME
(c) UJALA
(d) PM-KUSUM

Answers

Q106 — (b) Anaerobic digestion of organic matter

Q107 — (d) Methanogenesis — *Digestion ends in methanogenesis.*

Q108 — (c) 55–65% — *Biogas 55–65% CH_4.*

Q109 — (a) CO_2 (35–44%)

Q110 — (b) 35 °C — *Optimum ~35 °C.*

Q111 — (b) 20–30:1 — *C:N 20–30.*

Q112 — (b) Floating steel/FRP gas holder — *KVIC floating drum vs Deenbandhu fixed dome — the classic comparison.*

Q113 — (b) Constant gas pressure — *Floating drum gives constant pressure; the steel drum corrodes.*

Q114 — (b) Fixed-dome plant (no moving part, cheaper) — *Deenbandhu: no moving part, cheap.*

Q115 — (b) Variable gas pressure — *Fixed dome: variable pressure.*

Q116 — (b) Nitrogen-rich manure

Q123 — (c) 90% — *CBG IS 16087:2016 ≥ 90% methane.*

Q124 — (c) 250 bar — *Compressed to ~250 bar.*

Q125 — (a) SATAT — *SATAT scheme (as given in the source notes: 132 plants / 920 TPD commissioned, July 2026).*

Sources: Rai — Non-Conventional Energy Sources; Mittal — Energy in Agriculture; MNRE Biogas/SATAT programmes & PIB (as given in the source notes, July 2026); IS 16087:2016 (CBG specification); standard bioenergy texts; NPTEL bioenergy & energy-management lectures.